

1. Problem Statement

Organizations generate large amounts of employee data, but raw HR data often contains missing values, duplicate records, inconsistent department names, invalid values, and non-standard date formats. Without a structured analytical process, it becomes difficult to understand workforce distribution, salary patterns, hiring trends, and differences across departments.

The objective of this project is to transform raw employee data into a reliable analytical dataset and develop an interactive Power BI dashboard that helps HR and management understand workforce size, departmental distribution, salary levels, salary variation, and employee hiring trends.

2. Project Objectives

- Identify and resolve data-quality issues in the raw employee dataset.
- Create a clean and consistent employee dataset for analysis.
- Analyze employee distribution across departments.
- Analyze salary distribution and departmental salary differences.
- Identify employee hiring trends over time.
- Build an interactive Power BI dashboard with KPI cards, charts, filters, and a summary table.
- Generate business insights for HR reporting and workforce planning.

3. Dataset Information

The project started with a raw employee dataset containing 500 records and 7 core fields. After cleaning, the final analytical dataset contains 470 employee records.

Field	Description	Data Type
Emp_ID	Unique employee identifier	Integer
Full_Name	Employee full name	Text
Age	Employee age	Numeric
Join_Date	Employee joining date	Date
Department	Employee department	Categorical
Salary	Employee salary	Numeric
Email	Employee email address	Text

Raw records: 500 | Final analytical records: 470 | Records removed during duplicate handling: 30

4. Data Quality Assessment

The raw dataset contained missing values in Age, Join_Date, Department, Salary, and Email. It also contained duplicate records, inconsistent department labels, and invalid negative age values.

Issue in Raw Data	Observed	Treatment in Final Dataset
Total records	500	470 final analytical records
Duplicate records	30 duplicate rows	Removed
Missing Age	59	Resolved
Missing Join_Date	27	Resolved and standardized
Missing Department	17	Resolved and categorized

Missing Salary	36	Resolved
Missing Email	23	Resolved
Invalid negative ages	7 records	Corrected
Department naming inconsistency	Multiple variants/cases	Standardized into 7 categories

5. Data Cleaning Process

Duplicate Handling: Duplicate employee records were identified and removed, reducing the dataset from 500 to 470 records.

Missing-Value Handling: Missing values in key analytical fields were resolved so the final analytical dataset contains no null values.

Age Validation: Invalid negative age values were corrected; the final employee age range is 22–58 years.

Department Standardization: Department variants were consolidated into Sales, Operations, Finance, IT, Marketing, HR, and Other.

Date Standardization: Join dates were converted into a consistent date format for year-wise hiring analysis.

Salary Validation: Salary values were standardized as numeric values and checked for minimum and maximum consistency.

Final Validation: The cleaned dataset was checked for nulls, duplicates, invalid ages, and usable categorical values before dashboard creation.

6. Exploratory Data Analysis (EDA)

The cleaned dataset contains 470 employees with total salary expenditure of ₹48.71M and an average salary of ₹103,641.49.

Metric	Value
Total Employees	470
Total Salary	₹48.71M
Average Salary	₹103,641.49
Median Salary	₹102,000
Minimum Salary	₹25,000
Maximum Salary	₹180,000
Employee Age Range	22–58 years
Joining Period	2018–2025

6.1 Department Analysis

Department	Employees	Average Salary	Minimum Salary	Maximum Salary
Finance	77	₹107,655.84	₹28,000	₹174,000
HR	65	₹96,753.85	₹27,000	₹173,000
IT	76	₹103,934.21	₹25,000	₹179,000
Marketing	76	₹100,644.74	₹25,000	₹179,000
Operations	79	₹103,126.58	₹27,000	₹177,000
Other	17	₹130,823.53	₹33,000	₹172,000
Sales	80	₹102,675.00	₹31,000	₹180,000

6.2 Hiring Trend Analysis

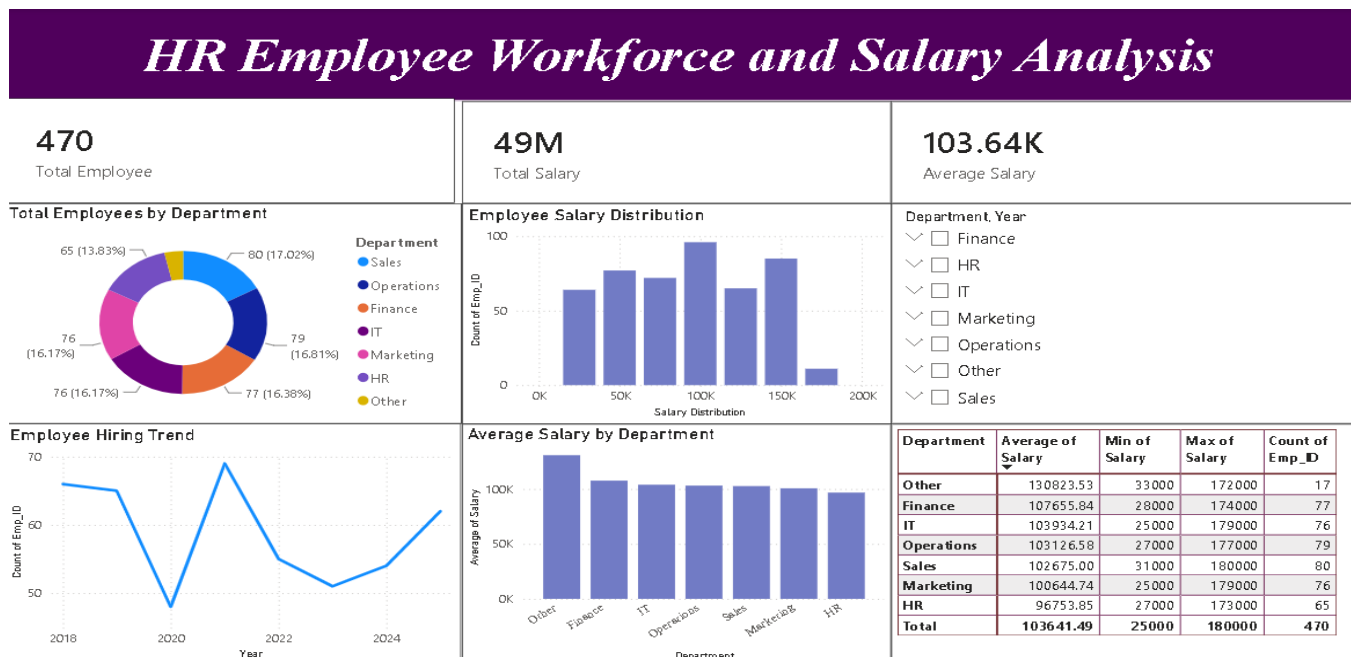
- 2018: 66 employees joined
- 2019: 65 employees joined
- 2020: 48 employees joined
- 2021: 69 employees joined
- 2022: 55 employees joined
- 2023: 51 employees joined
- 2024: 54 employees joined
- 2025: 62 employees joined

The highest number of employees joined in 2021 with 69 employees, while the lowest count in the observed period was 2020 with 48 employees.

7. Power BI Dashboard

The Power BI dashboard titled “HR Employee Workforce and Salary Analysis” provides an interactive HR view of workforce size, salary metrics, department distribution, salary distribution, hiring trends, and departmental salary statistics.

Dashboard Component	Purpose
Total Employee Card	Shows 470 total employees.
Total Salary Card	Shows approximately ₹49M total salary expenditure.
Average Salary Card	Shows approximately ₹103.64K average salary.
Employees by Department	Shows workforce distribution across seven departments.
Employee Salary Distribution	Shows employee count across salary ranges.
Employee Hiring Trend	Shows year-wise joining volume from 2018 to 2025.
Average Salary by Department	Compares average salary across departments.
Department Summary Table	Shows average, minimum, maximum salary and employee count.
Department / Year Filters	Allows interactive filtering by department and year.



8. Key Business Insights

- The cleaned dataset contains 470 employees with approximately ₹48.71M in total salary expenditure.
- Sales has the largest employee count at 80, followed by Operations at 79. HR has the smallest workforce among the named departments at 65.
- Finance has the highest average salary among the six named core departments at approximately ₹107.66K, while HR has the lowest at approximately ₹96.75K.
- The Other category has the highest average salary at approximately ₹130.82K, but it contains only 17 employees and should therefore be interpreted separately.
- Salary ranges from ₹25K to ₹180K, showing considerable variation in compensation levels.
- Hiring volume peaked at 69 employees in 2021 and was lowest at 48 employees in 2020.
- Employee counts across the major departments are relatively close, indicating a broadly distributed workforce.

9. Business Recommendations

1. Review compensation bands by department to understand whether salary differences align with job roles, experience, and market benchmarks.
2. Investigate the composition of the Other category because its high average salary may represent multiple job functions.
3. Use hiring trends for workforce planning and identify years with unusually high or low recruitment activity.
4. Monitor lower-headcount departments such as HR for possible capacity gaps during future hiring cycles.
5. Maintain a repeatable HR data-quality checklist covering duplicates, missing values, category consistency, dates, and salary validation.
6. Extend future versions of the dashboard with tenure, attrition, promotion, performance, or age-band analysis when relevant fields are available.

10. Conclusion

This project demonstrates an end-to-end HR analytics workflow in which raw employee data was assessed, cleaned, standardized, analyzed, and transformed into an interactive Power BI dashboard. The final dashboard provides a clear view of workforce size, departmental distribution, salary levels, salary variation, and hiring trends. These insights can support HR reporting, workforce planning, and compensation review.

11. Appendix – Project Summary

I worked on an HR Employee Workforce and Salary Analysis project using a raw dataset of 500 employee records. I performed data-quality checks, handled duplicates and missing values, standardized departments and dates, validated age and salary fields, and prepared a final analytical dataset of 470 records. I then performed exploratory analysis and developed a Power BI dashboard showing employee count, total and average salary, department-wise workforce distribution, salary distribution, hiring trends, and departmental salary statistics. The project converted raw HR data into management-friendly insights for workforce and salary analysis.